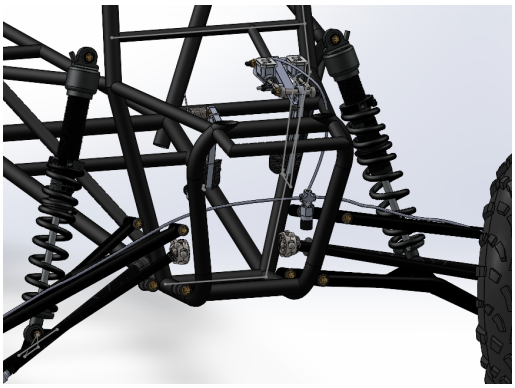
Cornell Baja Racing

Spring 2026

Technical Report



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Projects:
Brake Lines

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1 Brake Lines

1.1 Abstract

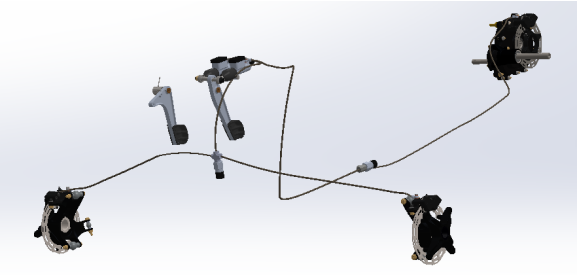


Figure 1: An illustration of how to put a figure into a LaTeX document.

Explanation The purpose of the brake lines is to carry the brake fluid from the master cylinders to the calipers. When the brake pedal is pressed the liquid pushes a piston that squeezes the calipers onto the rotor to make the wheel slow down.

1.2 Design Requirements

Explanation The lines must be able to follow the upper links in the front and the frame to the back of the car so that they go directly into the caliper without rubbing against the wheel and without risk of getting caught up in the drive shaft.

Constraints:

- Tubing cannot be plastic, and must be durable
- The lines must reach front and back lines with enough slack that the lines are never in tension.
- Must not fall below any portion of the car that is exposed to terrain.
- The lines must be attached to the frame tubes along its routing.
- Able to withstand the pressure of the brake fluid

Objectives:

- Minimize bumps in brake line routing
- leave space for rips due to small tolerances

1.3 Initial Research

1.3.1 Last Year's Car

Last year the X and T block shapes were adjusted to to be more square and weight saved. Due to the design of the masters and the reservoirs there was not a lot of space in front of the masters. The lines had one main flaw, which is the front lines routing had a large bump. The line had to go down one side and then up and over to the other side. This left a risk of bubbles at the top since any air that isn't bled out would rise and sit in that area. This would therefore affect the bleed of the brakes.

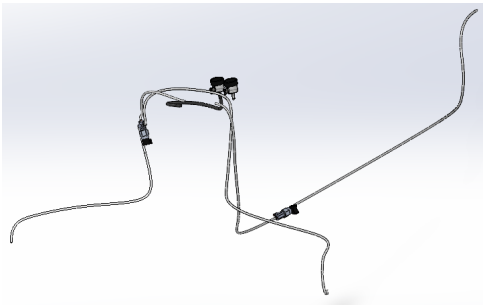


Figure 2: Example image of notable failure from previous year.

1.3.2 Another Older Car

TGXX had the original weight savings for the X and T blocks where the cut outs were added. It was very helpful to not weight the lines down and to also round them out a little bit. The issue with the routing was also present.

1.3.3 Online Research

Based on research, the best brake lines for the car would be the metal braided ones with an outer coating of plastic to make sure that the lines aren't easily damaged. This keeps them durable, while remaining flexible enough to do nice curves. The pressure sensors we have are also best positioned at the

direct impact of the line connecting to the masters (this doesn't matter as much).

1.4 High Level Description

1.4.1 X and T Blocks

The X and T blocks are the connectors between the line connecting to the master cylinders, the pressure sensor, and the remaining line(s). There are no notable changes to the blocks from last year. Each one has one multiple 6mm tapped holes for the lines and a larger 7/16 tapped hole for the pressure sensor. Each slide is faced down flat to ensure a tight seal. The lines and sensors are sealed to the blocks using copper crush washers.

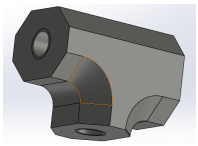


Figure 3: T block

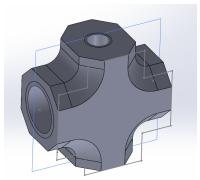


Figure 4: X block

1.4.2 Front Lines routing

The new masters gave a lot of space up front for the lines. This means that the front lines can now avoid going down on one side and then up again to get to the other side. The new route goes down the side near the brake pedal, gets to the x block, and branches out directly toward the calipers. The lines then go along the upper links.

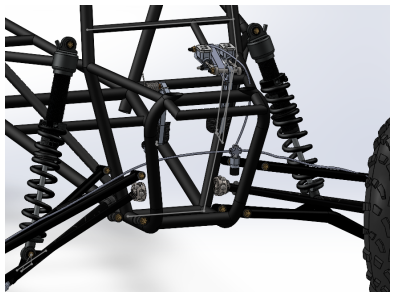


Figure 5: Front Lines

1.4.3 Rear Lines routing

The T block previously had an issue being too close to the front area and used to be facing downward rather than on its side. This had caused issues with getting in the way of the driver at times or being in the way during maintenance; for this the T block was moved farther down closer to the seat. The lines were then routed to move underneath the car along one of the connecting frame tubes, moving from one side to the other before connecting to the caliper.

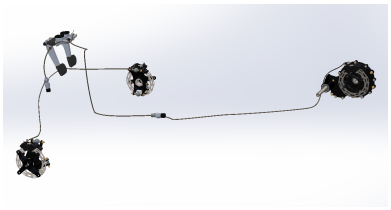


Figure 6: Rear lines in car space

1.4.4 Connection to the calipers

The calipers positioning was moved to the top of the rotor mount. This means that when the tires move while steering, the lines don't move as much. This means that the lines get to be tied tighter onto the frame and don't need as much slack, but also means that it is at higher risk of catching onto something around the wheels while the car drives.

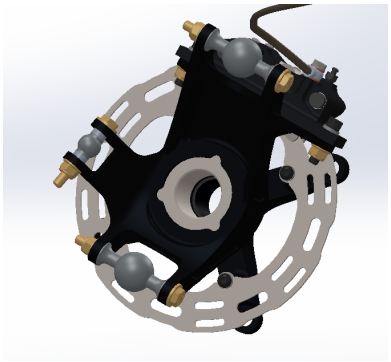


Figure 7: Rotor and Caliper View

1.5 Testing

1.5.1 Testing Procedure

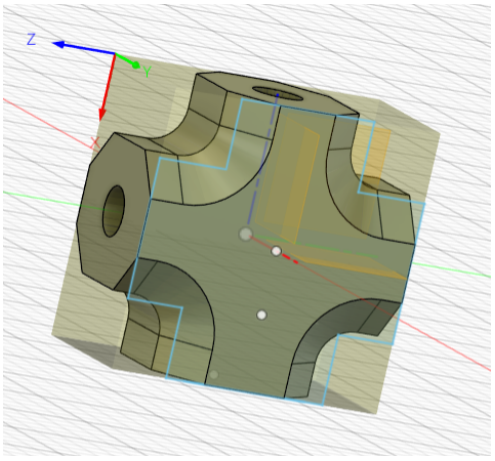
The procedure for testing is to have the lines and the rest of the brakes set up. Then the lines are bled to check if there are any leaks. Using a syringe at the calipers air and brake fluid is drawn out of the lines. If there is an excessive amount of air in the lines, there could be a leak in either the lines or where the lines connect. Another method we used to test it is setting up the syringe but also pressing on the brakes to push the fluid through. Once each test is finished we simply push the brake pedals to see if the caliper pistons are moving enough, which tells us if the lines are doing their job.

1.5.2 Conclusions

Near the end of testing we found little bubbles in the rear lines, and almost none in the front lines. Upon inspection there were no leaks in the lines. There was one spot near the right side caliper where the lines were damaged, but no harm came from the damage.

1.6 Manufacturing

The X and T blocks are machined using a CNC Tormach. Both blocks are made of small boxes of 6061 Aluminum. They machined by first cutting down the stock into a nice rectangle, and then drilling each hole with a different set up. The last operation is the cut outs which are there for weight saving.



As for the lines themselves they are bought and cut to size. Then they are fixed with the ends and tested to make sure they fit into the blocks.

1.7 Current Status

All the lines are currently complete, with two newly made x and t blocks as spares. There is enough length on the lines to remove any damage that is there.

1.8 Future Improvements

If I could redesign I would probably look for a new pressure sensor. We aren't quite sure whether the pressure sensors are causing any problems, but they are quite large; if there are smaller ones those would be nice. Additionally, I think it would be good to look into how to machine custom banjo bolts since they cost a decent bit to get on their own and they break easily.

2 Personal Reflection

My first year on the team was very eye opening. My previous experiences with engineering design were very limited and I had very little chances to manufacture. This year I was able to learn to machine in the shop and gain some tricks for the CNC. The part I was given had enough to do to fill the time but also left space for me to help other people out and learn about other parts. I believe the testing and assembly portions were the most fruitful for me as I got to learn about the rest of the car. I would really love to learn more about how to do calculations and analysis like Ansys, since that is not something I have gotten to touch yet. I am super excited to develop my skills next year.

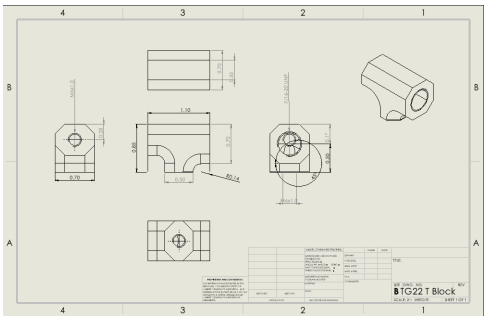
A Appendix

A.1 Part Summary

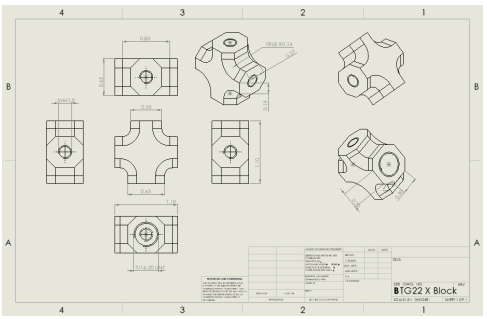
Brake Line Tube	Length
Masters to X	17in
X to longer side	35in
X to shorter side	26in
Masters to T	38in
T to rear	50in

Table 1: Summary of Brake Line Tube Lengths

A.2 Block' Part Drawings



T Block Drawing



X Block Drawing

A.3 References

- Zoe Szymanski's Spring 2023 Technical Report
- Dan O'Malley's Spring 2024 Technical Report